Datadog RUM Integration – Executive Summary

Project: DataPortal – Datadog Real User Monitoring (RUM) Integration
Objective: Enable business analytics by capturing user behavior events such as downloads, sessions, navigations, and user group activities.

Executive Summary

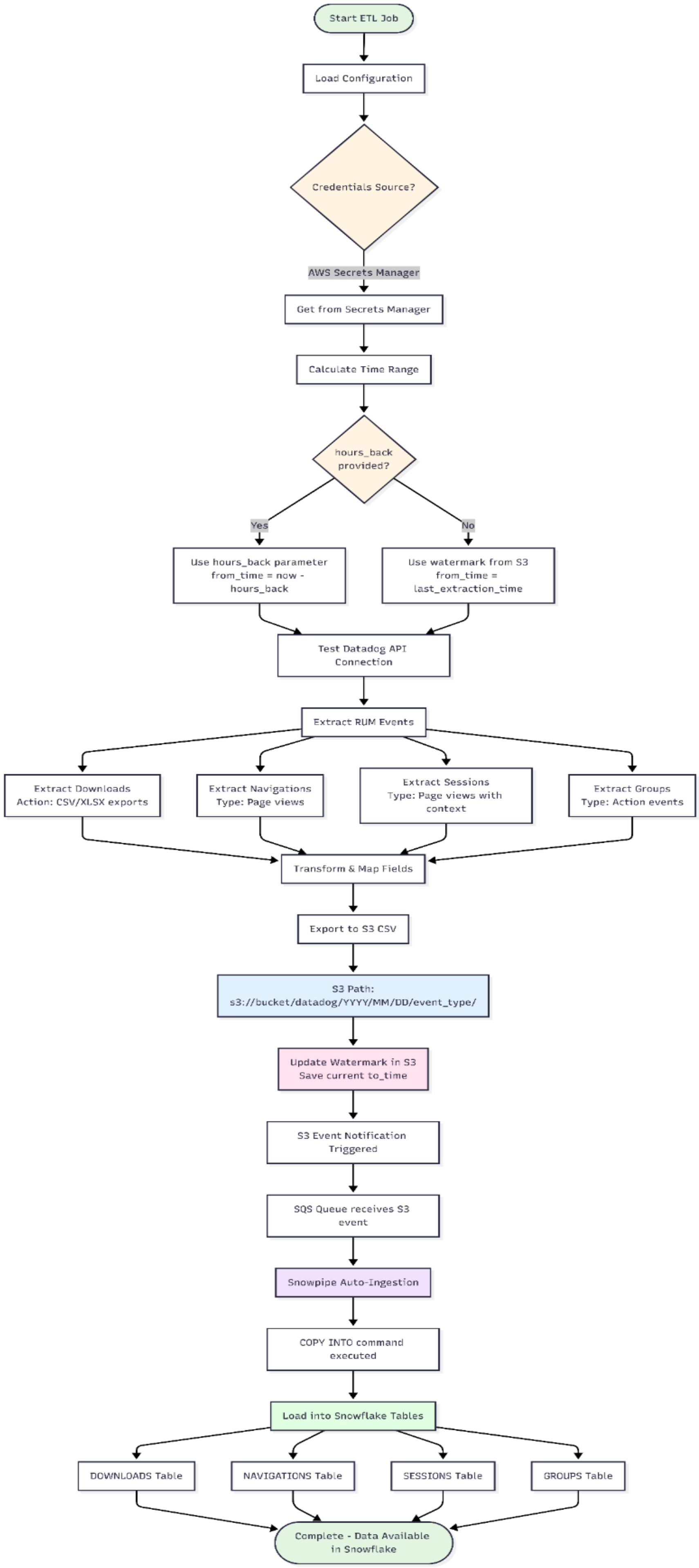
This solution implements a scalable and reliable data pipeline to extract Real User Monitoring (RUM) data from Datadog and load it into Snowflake for enterprise analytics. The design supports incremental data extraction using intelligent watermarking, configurable time-range selection, and standardized field mappings to ensure consistency and data quality.

Solution Overview

A Python-based ETL service queries the Datadog RUM API v2, processes multiple event types, and transforms the data into analytics-ready CSV files. The files are uploaded to Amazon S3 and automatically ingested into Snowflake using Snowpipe, enabling near real-time visibility into user behavior.

High-Level Architecture

The architecture below illustrates the end-to-end ETL workflow, including secure credential handling, incremental extraction logic, event processing, and automated data ingestion into Snowflake.


Start ETL Job
Load Configuration
Credentials Source?
AWS Secrets Manager
Get from Secrets Manager
Calculate Time Range
hours_back provided?
Yes
No
Use hours_back parameter from_time = now - hours_back
Use watermark from S3 from_time = last_extraction_time
Test Datadog API Connection
Extract RUM Events
Extract Downloads Action: CSV/XLSX exports
Extract Navigations Type: Page views
Extract Sessions Type: Page views with context
Extract Groups Type: Action events
Transform & Map Fields
Export to S3 CSV
S3 Path: s3://bucket/datadog/YYYY/MM/DD/event_type/
Update Watermark in S3 Save current to_time
S3 Event Notification Triggered
SQS Queue receives S3 event
Snowpipe Auto-Ingestion
COPY INTO command executed
Load into Snowflake Tables
DOWNLOADS Table
NAVIGATIONS Table
SESSIONS Table
GROUPS Table
Complete - Data Available in Snowflake

Key Highlights & Impact

- Implemented incremental extraction using S3-based watermarking to prevent data duplication.
- Enabled flexible backfill via configurable time windows (hours_back).
- Standardized event schemas across downloads, navigations, sessions, and group actions.
- Automated ingestion using S3 events, SQS, and Snowpipe for operational efficiency.
- Delivered analytics-ready datasets in Snowflake for product and business teams.